

MATH Unit 5 Question Bank — Graph Theory & Combinatorics

Mathematics (MATH) · Unit 5 · Graph Theory & Combinatorics · 2025

Important and frequently asked questions, grouped by marks.

Part A — Short answer (2 marks)

1. Compare Permutations and Combinations with Graph Colouring. When would you prefer one over the other?
2. Define Pigeonhole Principle. Derive or justify the key properties involved.
3. Explain Recurrence Relations with a neat diagram and a suitable example.
4. Explain Euler and Hamiltonian Paths with a neat diagram and a suitable example.
5. Write short notes on Graph Colouring.
6. What is meant by Planar Graphs? Discuss its advantages and limitations.

Part B — Long answer (10 marks)

1. Discuss the role of Permutations and Combinations in Graph Theory & Combinatorics.
2. Define Pigeonhole Principle. Derive or justify the key properties involved.
3. What is meant by Recurrence Relations? Discuss its advantages and limitations.
4. Write short notes on Euler and Hamiltonian Paths.
5. Compare Graph Colouring with Permutations and Combinations. When would you prefer one over the other?
6. Compare Planar Graphs with Recurrence Relations. When would you prefer one over the other?
7. Define Permutations and Combinations. Derive or justify the key properties involved.
8. Illustrate Pigeonhole Principle with a worked example.
9. Define Recurrence Relations. Derive or justify the key properties involved.
10. Discuss the role of Euler and Hamiltonian Paths in Graph Theory & Combinatorics.
11. What is meant by Graph Colouring? Discuss its advantages and limitations.
12. Illustrate Planar Graphs with a worked example.